Setup

8 networks x 3 images x 5 seeds, 16 channels, 2000 steps, 64x64 images, central 16x16 gap. Shared networks (1, 3, 5, 7) are scored at the best held-out step; unshared networks (2, 4, 6, 8) at the final step. All numbers are five-seed means from `results/comprehensive-target-aligned-report.md`.

Gap error as a multiple of grey (gap filled with 0.5)

Below 1 beats grey. Grey's gap MSE: gradient 0.005354, blurred noise (scale 4) 0.010777, stripes 0.125000.

Network	Sharing	Filters	Up/downsampling	Gradient	Blurred noise	Stripes
1	shared	3x3	on	0.004	0.10	0.38
3	shared	1x1	on	0.02	0.21	0.83
5	shared	3x3	off	0.23	0.33	0.49
7	shared	1x1	off	6.3	1.03	1.33
2	unshared	3x3	on	1.36	5.4	1.01
4	unshared	1x1	on	1.30	4.3	1.02
6	unshared	3x3	off	1.34	4.4	1.01
8	unshared	1x1	off	1.55	2.0	1.01
Harmonic fill				0.00	0.10	0.96

Correlation between gap output and true gap (Pearson r)

Network	Gradient	Blurred noise	Stripes
1	+0.999	+0.961	+0.805
3	+0.991	+0.930	+0.589
5	+0.928	+0.868	+0.736
7	-0.029	-0.008	-0.023
2, 4, 6, 8	-0.04 to +0.02	+0.01 to +0.05	-0.02 to +0.02
Harmonic fill	+1.000	+0.952	+0.272

Findings

1. No unshared network beats grey on any image, whatever its filter size or up/downsampling. Their gap output is uncorrelated with the true gap.
2. The unshared networks behave as "grey plus noise": gap output averages 0.5 and is unrelated to the image, so gap MSE = grey's MSE + variance of the gap output.
3. Harmonic fill is exact on the gradient and ties network 1 on blurred noise on average. Networks 1 and 5 clearly beat harmonic only on stripes.
4. Network 7 (shared, 1x1, no up/downsampling) has its best held-out step at step 50 in all 15 runs and gets worse with training.
5. Networks 1, 3 and 5 are mostly selected at steps 1650 to 2000 (network 5 at step 2000 in every blurred-noise run), so 2000 steps may be too few.

Provisional (not yet reproduced in this repo)

- The unshared networks' gap-output variance is about 30 times larger on blurred noise than on the other two images. 50-step trial runs (mostly one seed) indicate this depends on how far the visible pixel values sit from 0.5, not on image structure. Blurred noise is the only low-contrast image in the sweep, so for the unshared networks, differences across image families are dictated by contrast rather than image attributes.

Caveats

- One fixed blurred-noise image; seeds vary the network only.
- Shared and unshared networks use different checkpoint rules.
- Harmonic fill uses all visible pixels, including the 5% held out from training.